

1、 Environment Setup

This section uses **Ubuntu 22.04 + ROS Humble** as an example to explain how to set up and use a camera in a ROS 2 environment.

1、 Install Dependencies

Open a terminal and run:

```
sudo apt install libgflags-dev nlohmann-json3-dev libgoogle-glog-dev ros-humble-image-transport ros-humble-image-publisher
```

Install the libuvc library:

```
git clone https://github.com/libuvc/libuvc.git
cd libuvc
mkdir build && cd build
cmake .. && make -j4
sudo make install
sudo ldconfig # Refreshing the link library
```

2、 Compile

1) Create a Workspace

Using `orbbec_ws` as the workspace name under the `~` home directory:

```
mkdir orbbec_ws
cd orbbec_ws
mkdir src
```

2) Copy the Package into the Workspace

Extract the source archive, then copy the folders (packages) from the `src` directory into `~/orbbec_ws/src`.

3) Compile

Open a terminal and run:

```
cd ~/orbbec_ws
colcon build
```

4) Add Environment Variables

Open a terminal and run:

```
echo "source ~/orbbec_ws/install/setup.bash" >> ~/.bashrc
```

3、 Install udev Rules

Open a terminal and run:

```
cd orbbec_ws/src/OrbbecSDK_ROS/orbbec_camera/scripts/  
sudo sh install_udev_rules.sh
```

To verify that the rules file was loaded successfully and the camera is bound, run:

```
#gemini max  
ll /dev/gemini
```

```
sunrise@ubuntu:~$ ll /dev/gemini  
lrwxrwxrwx 1 root root 15 Nov 22 2023 /dev/gemini -> bus/usb/002/004  
sunrise@ubuntu:~$
```

If the output matches the screenshot above, the setup was successful.

4、 Launch the Camera and View Images

Open a terminal and run:

```
#gemini max camera start  
ros2 launch astra_camera gemini.launch.xml
```

```
Terminal - sunrise@ubuntu: ~
File Edit View Terminal Tabs Help
[astra_camera_node-1] height: 480
[astra_camera_node-1] fps: 30
[astra_camera_node-1] serial_number: ACRL53300J4
[astra_camera_node-1] format: mjpeg
[astra_camera_node-1] frame_id: camera_color_frame
[astra_camera_node-1] optical_frame_id : camera_color_optical_frame
[astra_camera_node-1] [INFO] [1716543117.567010825] [camera.camera]: open camera
  success
[astra_camera_node-1] [INFO] [1716543117.584074221] [camera.camera]: set depth v
ideo mode Resolution :640x480@30Hz
[astra_camera_node-1] format PIXEL_FORMAT_DEPTH_1_MM
[astra_camera_node-1] [INFO] [1716543117.584895317] [camera.camera]: set ir vide
o mode Resolution :640x480@30Hz
[astra_camera_node-1] format
[astra_camera_node-1] [WARN] [1716543117.585689997] [camera.camera]: Publishing
dynamic camera transforms (/tf) at 10 Hz
[astra_camera_node-1] [INFO] [1716543117.606410471] [camera.camera]: depth is st
arted
[astra_camera_node-1] [INFO] [1716543117.615371946] [camera.camera]: ir is start
ed
[astra_camera_node-1] [INFO] [1716543117.615528398] [camera.camera]: Start UVC c
amera
[astra_camera_node-1] [INFO] [1716543117.627042741] [camera.camera]: set uvc mod
e 640x480@30 format UVC_FRAME_FORMAT_MJPEG
[astra_camera_node-1] [INFO] [1716543118.308171997] [camera.camera]: device sta
rted.
```

To view the available topics, run:

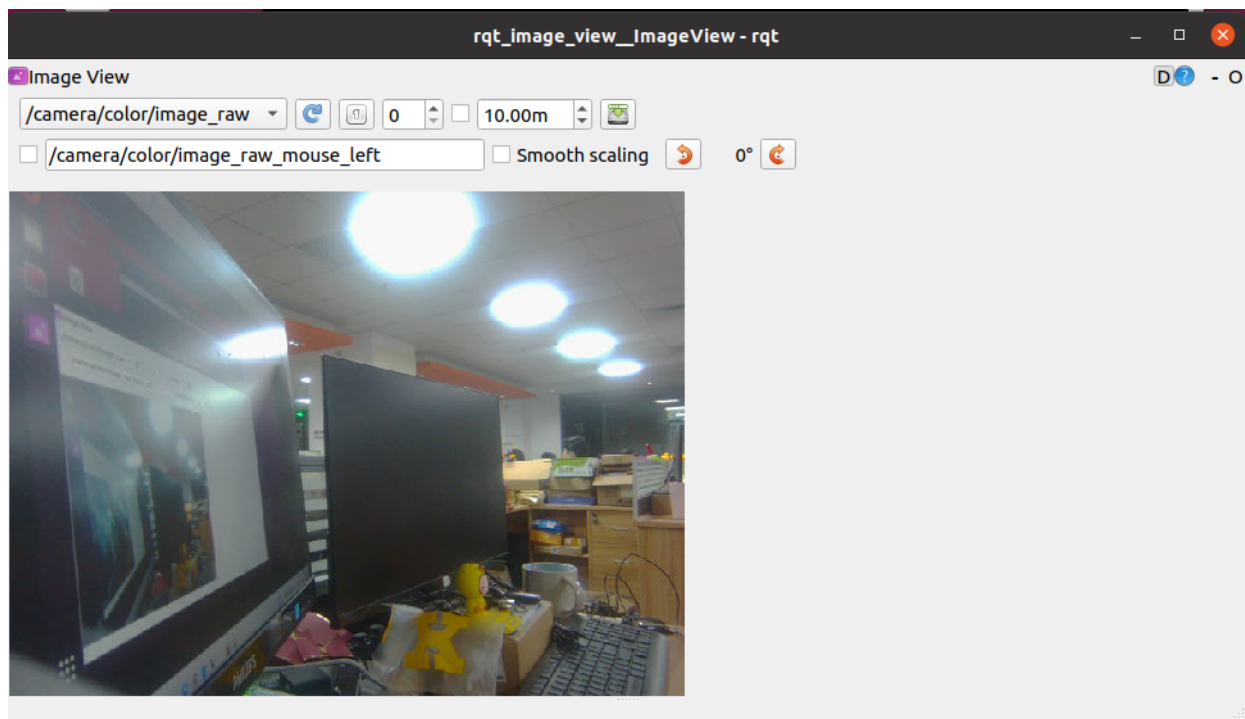
```
ros2 topic list
```

```
sunrise@ubuntu:~/orbbec_ws/src/yahboomcar_visual/simple_qrcode$ ros2 topic list
/camera/color/camera_info
/camera/color/image_raw
/camera/depth/camera_info
/camera/depth/image_raw
/camera/depth/points
/camera/ir/camera_info
/camera/ir/image_raw
/parameter_events
/rosout
/tf
/tf_static
```

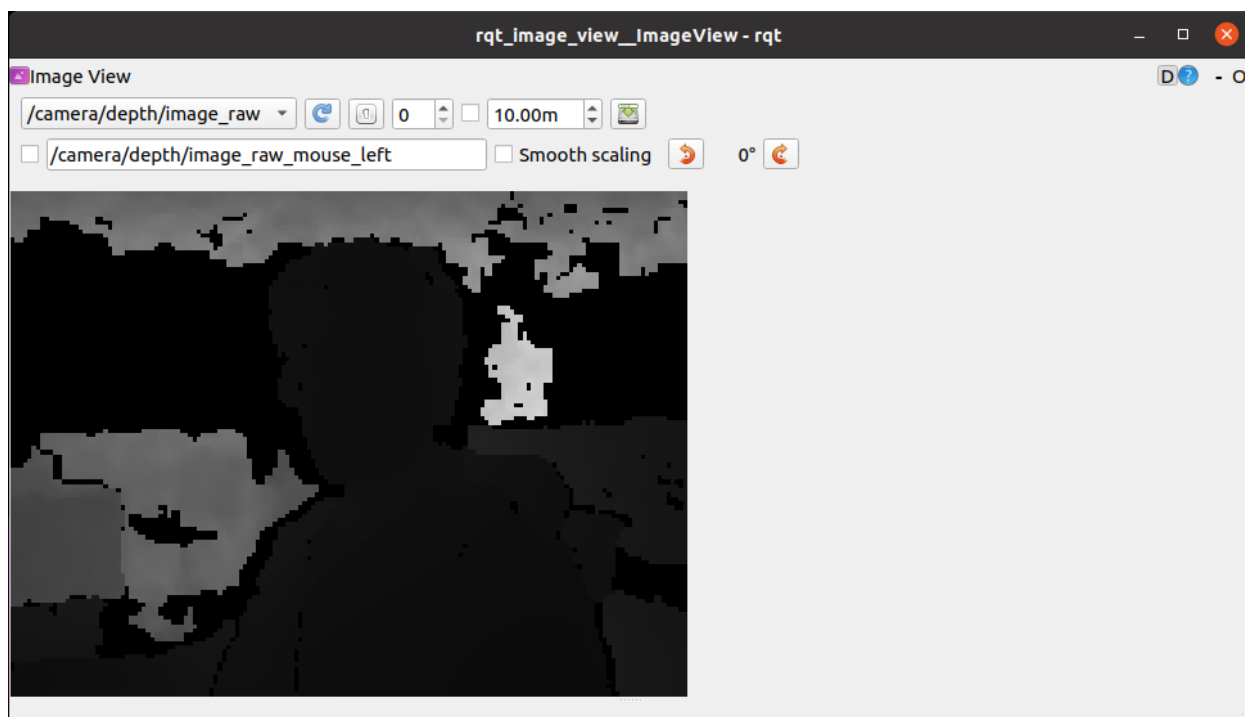
To view the camera images using the `rqt_image_view` tool, run:

```
ros2 run rqt_image_view rqt_image_view
```

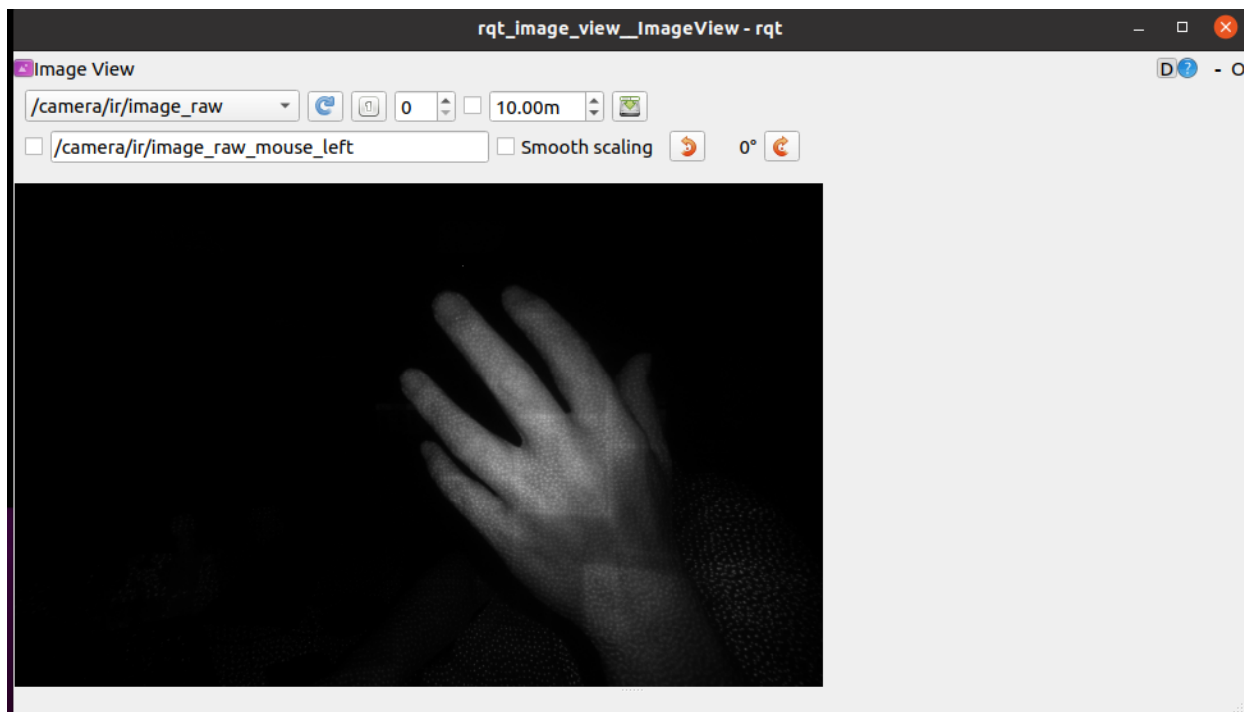
Color image



Depth image



Infrared (IR) image



In the rqt window, select the desired topic from the drop-down in the top-left corner to display.

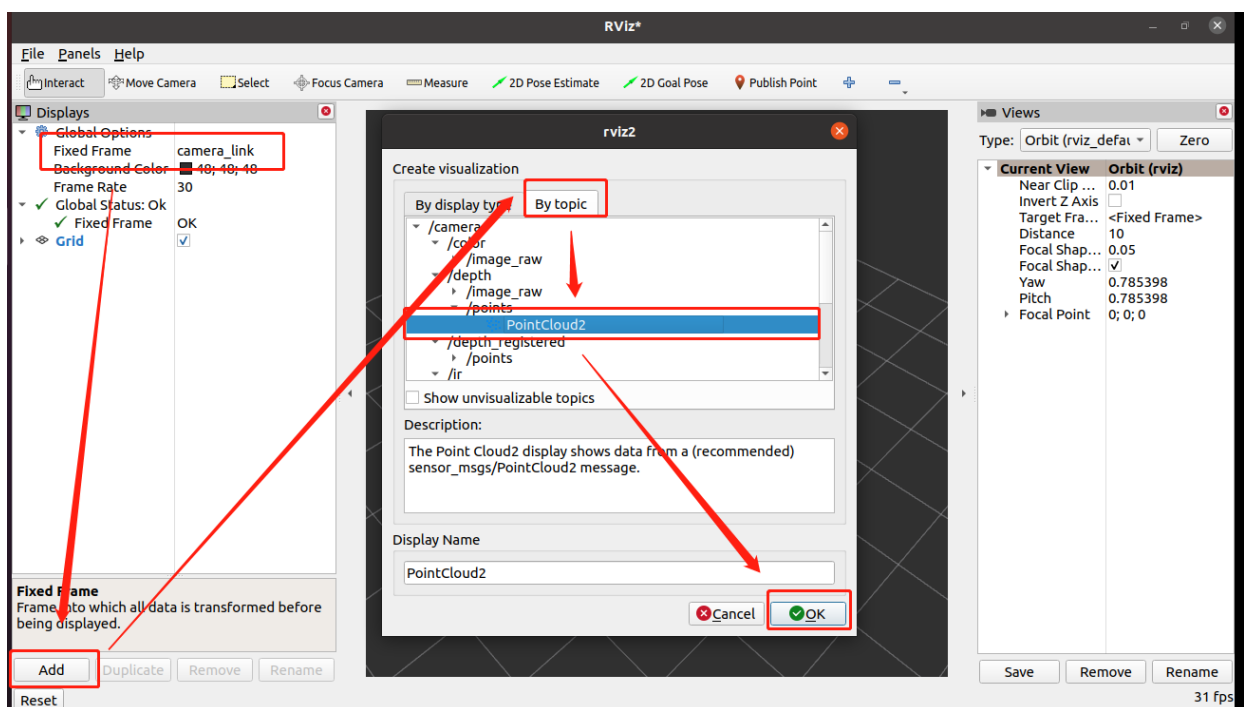
5、 Launch the Camera and View Point Cloud

Open a terminal and run:

```
ros2 launch astra_camera gemini.launch.xml
```

To visualize the point cloud data published by the camera, open RViz2:

```
rviz2
```



Once RViz2 is open, configure the point cloud visualization settings as shown in the screenshots above.

